

MTH253Z - Calculus: Sequences and Series – Syllabus

- **Credits:** 4
- **Instructor Name:** Dr. Radu Dascaluc
- **Instructor Email:** Radu.Dascaluc@oregonstate.edu
- **Office Hours:** Thursdays, 9:30-10:30 via Zoom or by appointment.

Course Description:

This course explores real-valued sequences and series, including power and Taylor series. Topics include convergence and divergence tests and applications. These topics will be explored graphically, numerically, and symbolically. This course emphasizes abstraction, problem-solving, reasoning, communication, connections with other disciplines, and the appropriate use of technology.

Prerequisites or Corequisites

MTH 252 or 252H. A minimum grade of C- is required in MTH 252 and MTH 252H.

Instructor Communication

Please post all course-related questions in the Q&A Discussion Forum so that the whole class may benefit from our conversation. Please contact me privately via OSU Email (preferred) or Canvas inbox for matters of a personal nature. You can expect a reply to your questions within 24 hours on business days (Monday-Friday). Grading and providing feedback on your assignments and activities may take up to five days. If I need more time, I will let you know when to expect a response.

Expectations for Time and Participation

This course combines approximately 120 total hours of instruction, online activities, and assignments for 4 credits spread over a 10-week period. Therefore, on average, you will spend approximately 12 hours per week on this course. This course is asynchronous and somewhat flexible, but not self-paced. Our schedule of Course Content and the due dates that appear in Canvas provide guidelines for how you'll interact and with what frequency. I recommend that you create your own workload schedule and set reminders for assignment due dates.

Learning Resources

Textbooks:

- OpenStax College. (2016). Calculus, Volume 2:

<https://openstax.org/details/books/calculus-volume-2?Book%20details> (CC-BY-NC-SA)

- OpenStax College. (2016). Calculus, Volume 1:

<https://openstax.org/details/books/calculus-volume-1?Book%20details> (CC-BY-NC-SA)

We will be using CANVAS for communication. Achieve is our required online homework platform, as well as Gradescope, and other internet applications. Information on these platforms is available on our course web page in CANVAS.

Technical Assistance

If you experience any errors or problems while in your online course, contact 24/7 Canvas Support through the Help link within Canvas. If you experience computer difficulties, need help downloading a browser or plug-in, or need assistance logging into a course, contact the IS Service Desk for assistance. You can call (541) 737-8787 or visit the [Service Desk](#) online.

Measurable Student Learning Outcomes

A successful student in MTH 253 will be able to:

1. Recognize and define sequences in a variety of forms and describe their properties, including the concepts of convergence and divergence, boundedness, and monotonicity.
2. Recognize and define series in terms of a sequence of partial sums and describe their properties, including convergence and divergence.
3. Recognize series as harmonic, geometric, telescoping, alternating, or p-series, and demonstrate whether they are absolutely convergent, conditionally convergent, or divergent, and find their sum if applicable.
4. Choose and apply the divergence, integral, comparison, limit comparison, alternating series, and ratio tests to determine the convergence or divergence of a series.
5. Determine the radius and interval of convergence of power series, and use Taylor series to represent, differentiate, and integrate functions.
6. Use techniques and properties of Taylor polynomials to approximate functions and analyze error.

Evaluation of Student Performance

At the end of the term your final grade will be based on the following categories and weights, calculated, and rounded to the nearest tenths place, and assigned a letter grade, according to the scale below.

Category	Percent
Online Homework (10)	20%
Written Homework (5)	15%
Discussion Boards (3)	10%

Unit Quizzes (4)	40%
Final Exam	15%
Total	100%

Letter Grade Scale

Letter Grade	Percent
A	100-92.5%
A-	92.49-89.5%
B+	89.49-86.5%
B 86.49-82.5% B-	82.49-79.5%
C+	79.49-76.5%
C 76.49-72.5% C-	72.49-69.5%
D+	69.49-66.5%
D 66.49-62.5%	62.49-59.5%
F	Below 59.5

Academic Calendar

All students are subject to the registration and refund deadlines as stated in the Academic Calendar: <https://registrar.oregonstate.edu/osu-academic-calendar>.

Course Policies

Discussion Participation

Students are expected to participate in all graded discussions. While there is great flexibility in online courses, this is not a self-paced course. You will need to participate in discussions on at least two different days each week, with your first post due no later than Tuesday evening of the discussion week; your second post is due by Wednesday; and the third post is due by the end of the week. Only one response post is due in the Week 1 Discussion board on Sunday.

Late Work Policy

The online homework assignments will allow you to submit your assignment up to three days late with a 10% deduction per day. The written homework assignments can be submitted up to two days late with a 10% deduction per day. The written assignments cannot be accepted beyond this point. Discussion boards cannot be submitted after the close of the unit for credit.

Use of AI Policy

For this course, you must be the sole author of all work. The use of any GenAI tools (including but not limited to ChatGPT, Copilot, or Claude) to generate or edit any portion of the substance of your work is not permitted for quizzes and exams or the written homework assignments.

In the process of studying and preparing for homework, quizzes, and exams you may use GenAI (such as ChatGPT, Copilot, or Claude) in some minor ways, for example while researching answers to your specific questions about the topics covered in class or getting help while stuck on a Web Homework assignment. However, such use must be combined with other type of research, such as studying the relevant material in the textbook and video lectures, as well as via participations in our Q&A discussion forums and asking questions during office hours. You must not use GenAI tools to solve your written or web-based homework problems for you.

Exam/Quiz Policies

There is no re-taking of exams/quizzes in this course. Once an exam/quiz has started, a student is no longer eligible for a make-up exam/quiz. If you need to miss an exam/quiz due to illness or an extenuating circumstance, a make-up exam/quiz may be granted. However, if you are unable to complete the make-up exam/quiz within three days of the scheduled exam, then your final exam grade will count in place of your missed exam grade.

Incompletes

When a requirement of a course has not been completed for reasons acceptable to the instructor and the rest of the academic work is passing, a report of "I" (incomplete) may be made and additional time granted, according to Academic Regulation 17 of [OSU Academic Regulations](https://regulations.oregonstate.edu/). In addition to this policy, to be eligible for an incomplete, students **MUST** have completed at least 70% of the coursework and be passing the course at the time of the incomplete. There will be no exceptions made to this policy. If you are having any difficulty that might prevent you completing the coursework, please don't wait until the end of the term; let me know right away.

Class Participation and Building Community

Active interaction with peers and your instructor is essential to everyone's success in this online course. I encourage you to please practice the following:

- Value the diversity of the class. Recognize and respect the experiences, abilities, and knowledge each person brings to our learning environment.
- Challenge others' ideas with the intent of facilitating growth. Acknowledge your peers' contributions and highlight areas of further inquiry.
- Be open to being challenged on your ideas or prejudices.
- Practice self-awareness in your communication with peers and consider that your comments may hurt others unintentionally.
- Assume the best of your classmates and instructor and expect the best from them.

Tutoring and Writing Assistance

You can connect live with experienced online tutors by accessing Online Tutoring in the side navigation bar of your Canvas course. You are eligible for up to 5 hours of tutoring each week. To learn more, go to [Online Tutoring - Overview](#).

Ecampus Reach Out for Success

University students encounter setbacks from time to time. If you encounter difficulties and need assistance, it's important to reach out. Consider discussing the situation with an instructor or academic advisor. Learn about resources that assist with wellness and academic success [here](#). You can also access the *Resources* tab in Canvas' global navigation menu for additional information.

Ecampus students are always encouraged to discuss issues that impact your academic success with the [Ecampus Success Team](#). Email ecampus.success@oregonstate.edu to identify strategies and resources that can support you in your educational goals.

For mental health:

Learn about [counseling and psychological resources](#) for Ecampus students. If you are in immediate crisis, please call or text the Suicide and Crisis Lifeline at 988 or Crisis Text Line by texting 741-741. You can also call CAPS at 541-737-2131 (after hours, select option 1).

For financial hardship:

Any student whose academic performance is impacted due to financial stress or the inability to afford groceries, housing, and other necessities for any reason is urged to contact the Director of Care for support (541-737-8748).

Expectations for Student Conduct

Student conduct is governed by the university's policies, as explained in the Student Conduct Code (<https://scs.oregonstate.edu/code-student-conduct>). Students are expected to conduct themselves in the course (e.g., on discussion boards, email postings) in compliance with the university's regulations regarding civility.

Academic Integrity

It is important that you understand what student actions are defined as academic misconduct at Oregon State University. The OSU Ecampus offers a [tutorial on academic misconduct](#), and you can also refer to the OSU [Student Code of Conduct](#) and the OSU [Student Community Standards](#) for more information. More importantly, if you are unsure if something will violate our academic integrity policy, ask your professors, GTAs, academic advisors, or academic integrity officers. Academic misconduct, or violations of academic integrity, can fall into seven broad areas, including but not limited to: cheating; plagiarism; falsification; assisting; tampering; multiple submissions of work; and unauthorized recording and use.

Statement Regarding Students with Disabilities

Accommodations for students with disabilities are determined and approved by Disability Access Services (DAS). If you, as a student, believe you are eligible for accommodations but have not obtained approval, please contact DAS immediately at 541-737-4098 or at <http://ds.oregonstate.edu>. DAS notifies students and faculty members of approved academic accommodations and coordinates implementation of those accommodations. While not required, students and faculty members are encouraged to discuss details of the implementation of individual accommodations.

Accessibility of Course Materials

All materials used in this course are accessible. If you require accommodations, please contact [Disability Access Services](#) (DAS). Additionally, Canvas, the learning management system through which this course is offered, provides a [vendor statement](#) certifying how the platform is accessible to students with disabilities.

Statement Regarding Religious Accommodation

Oregon State University is required to provide reasonable accommodations for employee and student sincerely held religious beliefs. It is incumbent on the student making the request to make the faculty member aware of the request as soon as possible prior to the need for the accommodation. See the [Religious Accommodation](#) Process for Students.

Student Bill of Rights

OSU has twelve established student rights. They include due process in all university disciplinary processes, an equal opportunity to learn, and grading in accordance with the course syllabus: <https://asosu.oregonstate.edu/advocacy/rights>.

Student Learning Experience Survey

During Fall, Winter, and Spring term the online Student Learning Experience surveys open to students the Wednesday of week 9 and close the Sunday before Finals Week. Students will receive notification, instructions, and the link through their ONID email. They may also log into the survey via MyOregonState or directly at <https://beav.es/Student-Learning-Survey>. Survey results are extremely important and are used to help improve courses and the learning experience of future students. Responses are anonymous (unless a student chooses to "sign" their comments, agreeing to relinquish anonymity of written comments) and are not available to instructors until

after grades have been posted. The results of scaled questions and signed comments go to both the instructor and their unit head/supervisor. Anonymous (unsigned) comments go to the instructor only.

Course Content & Schedule

Unit	Week	Assignments / Topic
1	1	Introduction Online Homework #1 Graded Discussion Board #1 Evaluating Limits Unit 1 Written Homework
1		Indeterminate forms L'Hospital's Rule Improper Integrals - Type I and Type II Unit 1 Quiz
2	2	Online Homework #2 Introduction to Sequences
2	3	Introduction to Series Online Homework #3 Unit 2 Written Homework Unit 2 Quiz
3	4	Online Homework #4 P-Series, Integral Test Error Analysis Approximation Test for Divergence
3	5	Online Homework #5 Comparison Test Limit Comparison Test Alternating Series Test
3	6	Graded Discussion Board #2 Ratio Test Online Homework #6 Unit 3 Written Homework

		Unit 3 Quiz Root Test
4	7	Online Homework #7 Introduction to Power Series
4	8	Online Homework #8 Unit 4 Written Homework Taylor and Maclaurin Series
4	9	Online Homework #9 Graded Discussion Board #3 Unit 4 Quiz Differentiation & Integration
5	10	Online Homework #10 Complex Series Euler Identities Unit 5 Written Homework Error Analysis Approximation Applications
Finals	Finals	Final Exam Covers all topics